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# Domain-Specific Retrieval-Augmented Generation for English-to-Hindi Machine Translation Using BLOOMZ-3B

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**Abstract**

In a linguistically diverse country like India, achieving high-quality English-to-Hindi translation across specialized domains is both crucial and complex. This work explores the power of Retrieval-Augmented Generation (RAG) frameworks, which enhance traditional machine translation by retrieving domain-relevant examples to guide the language model, thereby addressing challenges like data sparsity and contextual ambiguity. The proposed framework is a domain-adaptive RAG-based MT system using the BLOOMZ-3B model with FAISS-based retrieval employing three multilingual sentence embeddings: MiniLM, DistilUSE, and LaBSE for English to Hindi Translations. The evaluation spans five domains—Healthcare, Legal, Sports, Education, and News—assessing translation quality with multiple metrics. The proposed work demonstrates that the BLOOMZ-3B model with MiniLM embeddings consistently yields the highest BERT Similarity scores with 0.90 to 0.97, indicating semantically meaningful translations across domains. Compared to zero-shot approaches, the RAG system significantly reduces hallucinations and improves domain alignment. This study highlights the effectiveness of retrieval-augmented methods combined with strong semantic embeddings for enhancing domain-specific English-to-Hindi MT, with potential for extension to other languages and models.

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*Keywords:* Retrieval Augmented generation, Machine translation, FAISS, BLOOMZ, Large Language Model, Indic Languages

## 1. Introduction

Machine Translation (MT) is a fundamental task of Natural Language Processing (NLP), designed to automatically translate text or speech from one language to another. MT for Indian languages has become increasingly vital in bridging the linguistic diversity across the country's vast multilingual population. Despite advancements in neural MT as in [1], translating content across Indic languages remains a complex challenge due to linguistic richness, morphological complexity [2], and lack of high-quality parallel corpora. These challenges become even more pronounced in domain-specific translation, where accuracy, contextual precision, and terminology consistency are critical—especially in sensitive fields like healthcare, law, education, and news reporting. Generic MT systems often

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struggle to maintain semantic and syntactic integrity in such specialized scenarios. To address these limitations, Retrieval-Augmented Generation (RAG) has emerged as a effective solution. By combining the generative strength of Large Language Models (LLMs) with a retrieval mechanism that fetches relevant domain-specific contexts, RAG enables more grounded and factually accurate translations. Anju et al. [3] applied RAG with LLMs to support agricultural communication, showing domain-aware generation benefits. But proposed approach not only mitigates hallucinations common in zero-shot generation but also enhances domain adaptability without the need for extensive fine-tuning. While RAG has been successfully applied in tasks like question answering and summarization, its potential in Indic MT, particularly for domain-sensitive translation, remains underexplored.

The proposed research work presents a RAG based English-to-Hindi MT framework across various domains, using corpora from various domains. This framework is designed to overcome the limitations of general-purpose MT by leveraging retrieval-based contextualization, aiming to produce more accurate and coherent translations tailored to domain-specific requirements.
The key contribution of this work is given as follows:

- Proposed a domain-aware RAG-based English-to-Hindi translation framework that enhances contextual accuracy while significantly reducing the dependency on large parallel datasets by leveraging retrieval-based support.
- Integrate BLOOMZ-3B as the core language model, supported by multiple sentences embedding techniques for efficient retrieval.
- Utilize curated datasets from the Anuvaad corpus covering five key domains: Healthcare, Legal, Sports, Education, and News.
- Conduct a comparative study between RAG-based and zero-shot translation approaches across domains with multiple LLMs.

This research work explores various components of the research. Section 2 provides a review of existing literature related to the proposed work. In Section 3, the proposed model is outlined, with a detailed explanation of dataset, its architecture and components. Section 4 presents the study's results, including relevant statistical analyses. Section 5 concludes the paper by summarizing the findings and offering recommendations for future research directions. Lastly, Section 6 describes the challenges or limitations faced with the proposed framework.

## 2. Literature Survey

Recent research has made significant strides in improving machine translation for low-resource and Indian languages. For example, Nair et al. [4] showcased the effectiveness of BLOOMZ-3B in Indic language translation through prompt engineering and LoRA-based fine-tuning, while their follow-up study [5] evaluated LLMs on code-mixed Hindi-English translation. Gala et al. [6] introduced IndicTrans2, a comprehensive multilingual framework that offers high-quality translation support across all 22 scheduled Indian languages, significantly boosting accessibility. Mujadia et al. [7] systematically analyzed LLM performance on English-to-Indic language translation, highlighting both translation potential and persistent challenges. Zheng et al. [8] focused on domain-specific enhancements by fine-tuning LLMs with specialized terminologies, achieving better context relevance. Sani et al. [9] provided a wide-ranging survey on MT systems for Indian languages, detailing current limitations, available resources, and governmental initiatives.
Retrieval-Augmented Generation (RAG) has recently garnered attention for its ability to integrate external knowledge during generation, thus improving contextual grounding. Wang et al. [10] proposed a novel RAG-based MT framework that leverages unstructured knowledge to improve output quality, especially in knowledge-intensive tasks. Katsis et al. [11] introduced MTRAG, a multi-turn benchmark specifically designed to evaluate the conversational competence of RAG systems in dialogue-based scenarios. Vaishnav et al. [12] applied RAG for building conversational agents aimed at disseminating government welfare schemes in local languages, showcasing its social relevance. Darshan et al. [13] developed Chat2Fluency, an AI-driven language learning platform enhanced by RAG, emphasizing personalized education. Peri et al. [14] validated the use of a RAG-based chatbot for translating domain-specific medical texts, highlighting the benefits of retrieval in enhancing domain relevance.

Several studies have further stressed the importance of domain adaptation for effective translation in low-resource and culturally complex contexts. Mahapatra et al. [15] contributed MILPaC, a domain-specific legal benchmark tailored for Indian languages, which underscores the need for legal domain adaptation. Trivedi [16] focused on creating culturally adaptive conversational agents to support cross-lingual interaction in underserved languages. Dalal et al. [17] introduced AI Tutor, an educational tool that combines low-resource content with interactive AI to preserve and teach ancient Indian knowledge. Trivedi et al. [18] explored localization techniques for LLMs to ensure accurate translation that reflects cultural contexts. Finally, Guerreiro et al. [19] and Xu et al. [20] provided insights into hallucinations in neural MT, identifying them as a critical barrier in domain-specific applications where factual accuracy is paramount.

Despite significant advancements in machine translation and retrieval-augmented generation, several challenges remain. Most existing MT models for Indian languages struggle with domain-specific contexts, particularly in low-resource settings, leading to suboptimal translation quality. While RAG methods have shown promise in leveraging external knowledge, their application to domain-specific machine translation, especially for diverse and specialized fields like healthcare, legal, sports, education, and news, is limited. Moreover, there is a lack of comprehensive evaluations comparing RAG-based MT with zero-shot approaches across multiple domains using diverse metrics. Finally, issues like hallucination and efficiency in real world domain adaptation remain underexplored, indicating the need for targeted research that integrates retrieval-augmented techniques with domain-specific MT to enhance translation accuracy and reliability.

## 3. Methodology

The proposed methodology follows a structured pipeline wherein it begins with the retrieval of English-Hindi parallel corpora. Sentence embeddings are generated using a pre-trained encoder and indexed using FAISS for efficient retrieval. For each test input, the top-k most similar sentences are retrieved and used to construct an augmented prompt. This prompt is passed to the BLOOMZ-3B LLM, which generates the Hindi translation. The translations are then evaluated using a set of linguistic and semantic metrics to assess quality against ground truth references as shown in Fig 1.

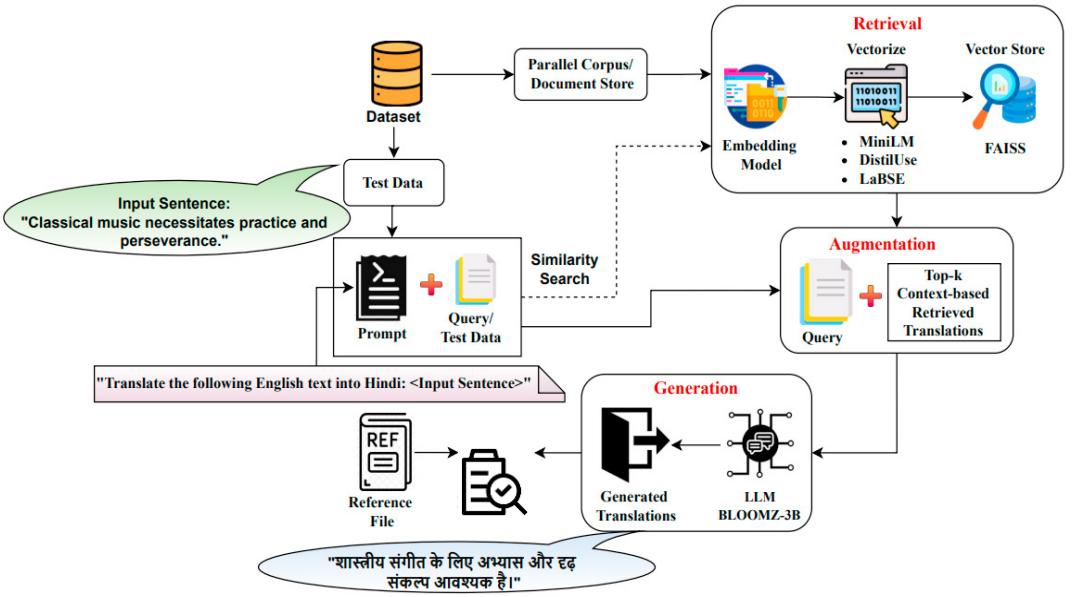



**Fig. 1.** Proposed architecture of RAG framework for English to Hindi Machine Translation

### 3.1. Document Store and Test Sets

The dataset used in the study is taken from the Anuvaad parallel corpus [21], publicly available on GitHub. This corpus provides English-Hindi parallel sentences across multiple domains, enabling domain-specific machine translation. The selected domains for this work include Healthcare, Legal, Sports, Education, and News. The RAG framework utilizes domain-specific corpora as Document Store and separate test sets for evaluation. Table 1 summarizes the number of instances considered for each domain in the retrieval corpora and test sets with sources. Due to computational resource constraints, the retrieval corpus for the Legal and News domains was down sampled from the original Anuvaad corpus, while the test sets for all domains except healthcare were also reduced in size for efficient evaluation.

**Table 1.** Size of the Parallel corpus of English to Hindi for each domain.

| Domain | Retrieval Corpus Size | Test Set Size | Sources |
|---|---|---|---|
| **Healthcare** | 1,761 | 441 | Lokmat |
| **Legal** | 1,00,000 | 1500 | Indian Judiciary |
| **Sports** | 59,412 | 1500 | Indian Express |
| **Education** | 15,118 | 1500 | Indian Express |
| **News** | 1,00,000 | 1500 | Indian express |

### 3.2. Retrieval module

The retrieval component is responsible for fetching relevant sentence pairs from the domain-specific parallel corpus to support the generation module in producing more accurate translations. It comprises two main components: a sentence embedding module and a vector database.

#### 3.2.1 Vector Database

For similarity-based retrieval, used FAISS (Facebook AI Similarity Search) [22], an open-source library optimized for high-speed vector similarity search, configured with cosine similarity as the distance metric across all experiments. It is used as a high-performance vector index for nearest neighbor search on dense sentence embeddings. In this implementation, the IndexFlatIP index is employed, which performs similarity search using the inner product (dot product). To enable cosine similarity search, sentence embeddings are normalized with L2 normalization before indexing, so that inner product similarity corresponds directly to cosine similarity. The FAISS index runs on CPU to enable compatibility with systems lacking GPU support, trading off some speed for broader accessibility and lower resource requirements.

#### 3.2.2 Sentence Embedding

Three different sentence embedding models generate dense vector representations of English sentences for retrieval-

- *MiniLM:* A compact transformer-based model producing 384-dimensional embeddings optimized for speed and efficiency. It supports multiple languages but is less explicitly multilingual than LaBSE or DistilUSE, making it a good tradeoff between performance and resource usage [27].
- *DistilUSE (distiluse-base-multilingual-cased-v2):* A distilled version of the Universal Sentence Encoder producing 512-dimensional embeddings. It supports over 15 languages including English and Hindi, enabling effective multilingual semantic understanding with relatively low computational cost [28].
- *LaBSE (Language-agnostic BERT Sentence Embedding):* A BERT-based model generating 768-dimensional embeddings trained for multilingual sentence representation across 100+ languages. Performs well in cross-lingual retrieval but demands higher memory and compute than smaller models. [28].

All embeddings are computed using available hardware acceleration where possible, normalized for cosine similarity, and indexed with FAISS CPU for scalable and efficient retrieval. This combination balances high-quality multilingual sentence representations with practical indexing for retrieval-augmented translation systems.

### 3.3. Augmentation Module

The augmentation module uses each test sentence as a query to retrieve the top-k most semantically similar sentences from the indexed training dataset using FAISS with cosine similarity. The test sentence is first converted into a dense vector embedding via the chosen sentence encoder, which is then used to search the FAISS index for the closest matches. The retrieved English sentences and their corresponding Hindi translations are incorporated into a prompt that begins with the instruction "Translate the following English text to Hindi:<input sentence>", followed by the retrieved sentence pairs, and concludes with the test English sentence. This structured prompt is provided to the language model to guide it in generating a relevant Hindi translation by leveraging contextual examples retrieved dynamically based on the test input. Table 2 summarizes the key RAG parameters used for translation, including retrieval settings and generation controls. While some parameters were tuned, others like top_p and repetition_penalty followed default values to maintain stable output quality.

**Table 2.** RAG Configuration Parameters for Indic Machine Translation

| Parameter | Value |
|---|---|
| **Number of Retrieved (Top-k)** | 5 |
| **Retrieval Chunk Size** | Sentence-level |
| **Vector Dimension** | 768 (LaBSE), 384 (MiniLM), 512 (DistilUSE) |
| **Scoring Metric for Retrieval** | Cosine Similarity or Dot product |
| **Temperature** | 0.7 |
| **Top-p (Nucleus Sampling)** | 0.9 |
| **Top-k (Token Sampling)** | 50 |
| **Repetition Penalty** | 1.1 |
| **Max Tokens Generated** | 256 |

### 3.4. Generation Module

The generation module utilizes the BLOOMZ-3B LLM [7], a multilingual, instruction-tuned variant of the BLOOM family with approximately 3 billion parameters. BLOOMZ-3B supports over 12 Indic languages, including Hindi and English, enabling robust cross-lingual generation tasks such as machine translation. The model architecture consists of 32 transformer layers, with a hidden size of 2560 and 20 attention heads, striking a balance between performance and computational efficiency. For translation, the model is prompted with a constructed input that includes the original English sentence along with examples retrieved from similar sentences, which helps the model produce more accurate and relevant translations. The decoding process uses parameters such as a temperature of 0.7 to control creativity and diversity, a maximum token length to limit output size, and padding with the model's EOS token to properly terminate sequences. This setup facilitates effective and corresponding generation of Hindi translations from English inputs.

### 3.5. Evaluation Module

Due to the inherent ambiguity and variability in language, multiple evaluation metrics were used to capture different facets of translation quality, including lexical, syntactic, and semantic accuracy. Metrics like BLEU, CHRF++, and METEOR (ranging from 0 to 100 for BLEU and CHRF++, and 0 to 1 for METEOR) focus mainly on lexical and syntactic overlap with the reference. COMET (ranging from -1 to 1) is a neural metric that relies on reference translations to evaluate the overall semantic adequacy and translation quality. In contrast, BERT Similarity (ranging from 0 to 1) leverages contextual embeddings to measure semantic closeness between the generated and reference sentences, capturing deeper meaning with less reliance on exact reference matching.

## 4. Results and Analysis

This section provides a comprehensive performance analysis of the results obtained from experiments and sub-divided into the following categories,

- Performance evaluation of RAG based Machine Translation
- Performance Evaluation of Zero-Shot Translation with multiple LLMs in comparision with the Proposed RAG Framework
- Performance evaluation with Few Samples across different approaches

### 4.1. Performance evaluation of RAG based Machine Translation

The RAG-based translation performance across domains using different sentence embeddings. Intially, using the combination of BLOOMZ-3B as the LLM and MiniLM as the sentence encoder, the system demonstrated strong translation performance across all domains. As seen in Table 3, This Combination achieved the highest scores overall, especially in domains like Education and Healthcare, with BLEU scores of 12.26 and 7.99, and BERT Similarity reaching up to 0.9653. The CHRF++ and METEOR scores were also consistently higher, indicating that the retrieved examples contributed significantly to the model's ability to generate contextually accurate and fluent Hindi translations.

**Table 3.** BLOOMZ-3B Translation Quality of Proposed RAG framework on each Domain

| DOMAIN | BLEU | CHRF++ | METEOR | COMET | Bert Similarity |
|---|---|---|---|---|---|
| | **Sentence Encoder – MiniLM** | | | | |
| **Healthcare** | 7.9923 | 31.5017 | 0.2613 | 0.7088 | 0.9653 |
| **Legal** | 9.5255 | 31.4682 | 0.2702 | 0.5543 | 0.9495 |
| **Sports** | 7.9111 | 30.9331 | 0.2413 | 0.6162 | 0.9340 |
| **Education** | 12.2664 | 32.1593 | 0.2636 | 0.6302 | 0.9096 |
| **News** | 9.1652 | 34.1874 | 0.2727 | 0.6358 | 0.9553 |
| | **Sentence Encoder – DistilUse** | | | | |
| **Healthcare** | 7.1072 | 29.8887 | 0.2472 | 0.6169 | 0.9606 |
| **Legal** | 9.3439 | 29.7358 | 0.2607 | 0.5347 | 0.8971 |
| **Sports** | 7.0308 | 29.4548 | 0.2283 | 0.5900 | 0.9055 |
| **Education** | 10.7967 | 31.1415 | 0.2528 | 0.6180 | 0.9062 |
| **News** | 8.2815 | 33.2676 | 0.2633 | 0.6139 | 0.9448 |
| | **Sentence Encoder – LaBSE** | | | | |
| **Healthcare** | 5.7159 | 24.3832 | 0.1926 | 0.5619 | 0.9578 |
| **Legal** | 4.9711 | 23.7018 | 0.1739 | 0.4947 | 0.9506 |
| **Sports** | 23.0277 | 36.5147 | 0.3281 | 0.6290 | 0.9633 |
| **Education** | 9.0932 | 25.5274 | 0.2042 | 0.6063 | 0.9374 |
| **News** | 4.7279 | 21.8917 | 0.1564 | 0.5467 | 0.9598 |

In the case of BLOOMZ-3B with DistilUSE as the encoder, the translation performance remained stable but slightly lower than that of MiniLM as shown in Table 3. The BLEU scores ranged between 7.03 and 10.79 across domains, and the COMET and BERT Similarity metrics also showed reasonable alignment with the ground truth. While this configuration maintained a balance between precision and semantic adequacy, it did not outperform MiniLM in any individual domain, making it a moderately effective choice.

When paired with LaBSE, BLOOMZ-3B produced mixed results. Although this combination yielded the highest BLEU score of 23.02 in the Sports domain, its performance dropped significantly in other domains such as News and Legal. The BLEU scores in these areas were as low as 4.72 and 4.97, and METEOR and CHRF++ values followed a similar declining trend as in Table 3. This indicates that while LaBSE can capture certain domain-specific similarities well, its retrieval quality lacks consistency across varied contexts.

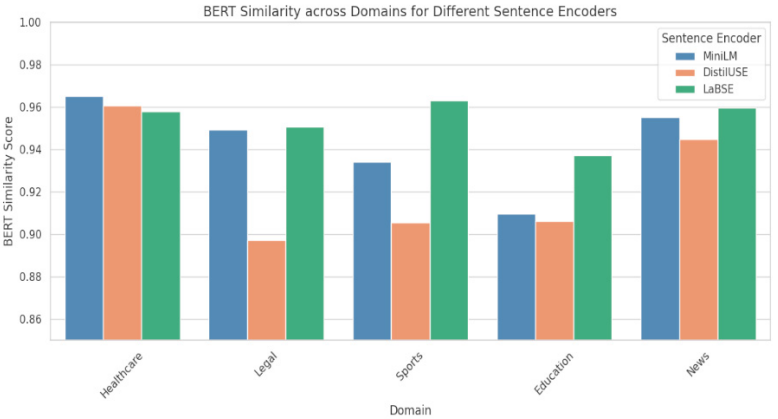



**Fig. 2.** Comparison of BERT Similarity Scores Across Domains with Sentence Encoders

The plot in Fig. 2 compares BERT similarity scores across various domains using different sentence encoders combined with the BLOOMZ-3B language model. Among the encoders, MiniLM consistently demonstrates the highest semantic alignment across all domains, making it the most effective overall. DistilUSE shows moderate and relatively stable performance, while LaBSE exhibits noticeable inconsistencies, particularly in domains like legal and news. Overall, the MiniLM and healthcare combination performs best, whereas LaBSE tends to underperform across most domains, indicating it may be less suitable for domain-specific semantic evaluations in this setup.

### 4.2 Performance Evaluation of Zero-Shot Translation with multiple LLMs in comparison with the Proposed RAG Framework

Baseline RAG implementations for Indic languages remain limited, particularly for Hindi, which constrains the availability of direct comparisons. Therefore, a zero-shot evaluation with three different baseline models such as BLOOMZ-3B, Airavata [29], a Hindi instruction tuned LLM and Llama-3.2-3B [30] were employed across all considered domains, serving as a consistent benchmark to assess the effectiveness of the proposed framework.

In the Zero-shot translation using LLMs without any retrieval-based augmentation, Baseline Models achieved reasonable performance across domains as shown in Table 4, with BLEU scores ranging from 7.7 to 31.4 and CHRF++ up to 55.4. Despite these variations, BERT Similarity remained consistently high (mostly above 0.90), showing that semantic meaning was generally preserved. However, the lack of retrieval support often led to weaker contextual grounding, particularly in domain-sensitive settings such as Legal and Education, where translations deviated from precise terminology. In contrast, the proposed RAG framework (BLOOMZ-3B + FAISS + LaBSE) delivered more context-aware translations, with strong COMET and BERT scores reflecting deeper semantic alignment, and a notable BLEU gain in the Sports domain (23.02 compared to ~8.0 in zero-shot). Importantly, hallucinations—less prominent in RAG but evident in zero-shot outputs, were also identified and analysed in Table 6, reinforcing the importance of retrieval in controlling translation reliability. This comparison highlights that while zero-shot models capture broad semantic structures, RAG ensures domain relevance, reduces hallucinations, and provides more coherent translations.

### 4.3. Performance evaluation with Few Samples across different Approaches

The domain-wise evaluation revealed that the combination of BLOOMZ-3B with FAISS-based retrieval and the LaBSE sentence encoder consistently achieved the highest BERT similarity scores across multiple domains as shown in Table 5. This indicates its strong ability to preserve the semantic integrity of translations. Based on this observation, generated representative translation samples using this specific configuration to highlight the model's effectiveness across various real-world domains.

**Table. 4.** Translation Quality of Zero Shot Approach on each Domain on various LLMs

| MODEL | DOMAIN | BLEU | CHRF++ | METEOR | COMET | Bert Similarity |
|---|---|---|---|---|---|---|
| BLOOMZ – 3B | Healthcare | 7.7166 | 30.3678 | 0.2505 | 0.6381 | 0.9514 |
| | Legal | 9.3673 | 28.6270 | 0.2487 | 0.5433 | 0.8574 |
| | Sports | 8.0071 | 30.4472 | 0.2308 | 0.6069 | 0.9275 |
| | Education | 9.7371 | 27.9622 | 0.2229 | 0.5941 | 0.8105 |
| | News | 8.6019 | 33.5838 | 0.2614 | 0.6313 | 0.9464 |
| Airavata | Healthcare | 15.7505 | 37.6209 | 0.3383 | 0.6760 | 0.9306 |
| | Legal | 7.4443 | 57.5199 | 0.5868 | 0.6877 | 0.9336 |
| | Sports | 8.2404 | 43.4940 | 0.3700 | 0.6665 | 0.9490 |
| | Education | 9.3930 | 41.0706 | 0.3459 | 0.6824 | 0.9475 |
| | News | 10.2278 | 47.1572 | 0.4087 | 0.6998 | 0.9679 |
| Llama – 3.2 – 3B | Healthcare | 9.0269 | 32.9715 | 0.2638 | 0.6208 | 0.9690 |
| | Legal | 20.0693 | 41.7636 | 0.4140 | 0.5936 | 0.9643 |
| | Sports | 10.0864 | 31.444 | 0.0345 | 0.3611 | 0.9126 |
| | Education | 15.6099 | 37.9333 | 0.3083 | 0.6541 | 0.9497 |
| | News | 11.6599 | 39.3613 | 0.3227 | 0.6181 | 0.9623 |
| Proposed Best RAG Model (BLOOMZ-3B + FAISS + LaBSE) | Healthcare | 5.7159 | 24.3832 | 0.1926 | 0.5619 | 0.9578 |
| | Legal | 4.9711 | 23.7018 | 0.1739 | 0.4947 | 0.9506 |
| | Sports | 23.0277 | 36.5147 | 0.3281 | 0.6290 | 0.9633 |
| | Education | 9.0932 | 25.5274 | 0.2042 | 0.6063 | 0.9374 |
| | News | 4.7279 | 21.8917 | 0.1564 | 0.5467 | 0.9598 |

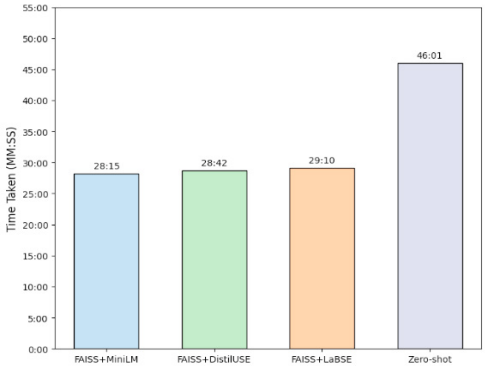



**Fig. 3.** Average Translation Time for RAG vs Zero-shot using BLOOMZ-3B Model (Test set size = 1500)

To conclude the results section, it was observed that while hallucinations were more frequent in Zero-shot translations of BLOOMZ-3B as shown in Table 6, due to the lack of domain-specific context, the overall semantic quality remained reasonably strong, with BERT Similarity scores staying consistently high. However, surface-level metrics like BLEU were lower, reflecting reduced lexical overlap. Table 6 also highlights how the RAG-based outputs provided more accurate and context-grounded translations, effectively avoiding hallucinations by leveraging retrieved domain-relevant examples. Although Zero-shot translations were not entirely poor, they took comparatively more time, as shown in Fig. 3, and lacked the contextual richness offered by the RAG framework, making the latter more effective for domain-specific translation.

**Table. 5.** Sample Translations of BLOOMZ-3B + FAISS + LaBSE and Zero shot Translations across Different Domains

| | | | |
|---|---|---|---|
| **Healthcare** | **Original Sentence** | The doctor advised her to stay hydrated and rest for a few days. | |
| | **Reference Sentence** | डॉक्टर ने उसे कुछ दिनों के लिए आराम करने और हाइड्रेटेड रहने को कहा। | |
| | **RAG Translation** | डॉक्टर ने उसे कुछ दिन आराम करने और पर्याप्त पानी पीने की सलाह दी। | |
| | **Scores** | **Bleu:** 81.3 | **Bert:** 0.96 |
| | **Zero Shot Translation** | डॉक्टर ने कहा कि वह पानी पीए और कुछ दिन सोए। | |
| | **Scores** | **Bleu:** 45.2 | **Bert:** 0.84 |
| **Legal** | **Original Sentence** | The petitioner argued that the notice issued was in violation of Article 14. | |
| | **Reference Sentence** | याचिकाकर्ता ने कहा कि यह नोटिस अनुच्छेद 14 के खिलाफ है। | |
| | **RAG Translation** | याचिकाकर्ता का कहना था कि यह नोटिस अनुच्छेद 14 का उल्लंघन करता है। | |
| | **Scores** | **Bleu:** 61.6 | **Bert:** 0.94 |
| | **Zero Shot Translation** | याचिकाकर्ता ने तर्क दिया कि जारी की गई नोटिस अनुच्छेद 14 का उल्लंघन थी। | |
| | **Scores** | **Bleu:** 39.5 | **Bert:** 0.80 |
| **Sports** | **Original Sentence** | The team failed to convert chances in the second half, leading to a draw. | |
| | **Reference Sentence** | टीम दूसरे हाफ में अवसरों को नहीं बदल पाई और मैच बराबरी पर समाप्त हुआ। | |
| | **RAG Translation** | दूसरे हाफ में टीम ने अवसरों को नहीं बदला और मैच ड्रॉ हो गया। | |
| | **Scores** | **Bleu:** 85.4 | **Bert:** 0.93 |
| | **Zero Shot Translation** | टीम ने सेकंड हाफ में मौके को नहीं किया, और ड्रॉ हुआ। | |
| | **Scores** | **Bleu:** 36.1 | **Bert:** 0.78 |
| **Education** | **Original Sentence** | Online learning platforms have grown rapidly after the pandemic. | |
| | **Reference Sentence** | महामारी के बाद ऑनलाइन लर्निंग प्लेटफॉर्म तेजी से बढ़े हैं। | |
| | **RAG Translation** | महामारी के बाद ऑनलाइन लर्निंग प्लेटफॉर्म्स में तेज़ वृद्धि देखी गई है। | |
| | **Scores** | **Bleu:** 91.1 | **Bert:** 0.90 |
| | **Zero Shot Translation** | महामारी बाद ऑनलाइन पढ़ाई साइट्स ज्यादा हो गया। | |
| | **Scores** | **Bleu:** 32.8 | **Bert:** 0.75 |
| **News** | **Original Sentence** | The Prime Minister addressed the nation during the emergency session. | |
| | **Reference Sentence** | प्रधानमंत्री ने आपात बैठक के दौरान राष्ट्र को संबोधित किया। | |
| | **RAG Translation** | आपातकालीन सत्र के दौरान प्रधानमंत्री ने देश को संबोधित किया। | |
| | **Scores** | **Bleu:** 65.3 | **Bert:** 0.95 |
| | **Zero Shot Translation** | प्रधानमंत्री ने देश से बोला इमरजेंसी मीटिंग में। | |
| | **Scores** | **Bleu:** 35.0 | **Bert:** 0.76 |

## 5. Conclusion and Future Scope

Machine Translation remains an essential tool for bridging linguistic divides in multilingual societies like India. This study introduced a domain-specific English-to-Hindi translation framework using Retrieval-Augmented Generation (RAG) with the BLOOMZ-3B language model. By employing FAISS-based dense retrieval and sentence encoders such as MiniLM, DistilUSE, and LaBSE, the system effectively retrieved contextually relevant examples to guide translation. Empirical evaluations across five diverse domains—Healthcare, Legal, Sports, Education, and News—revealed that the RAG-based approach consistently outperformed Zero-shot methods, particularly in semantic accuracy as reflected by higher BERT Similarity and COMET scores. While character-level metrics like BLEU and METEOR were slightly lower due to flexible paraphrasing, their impact on meaning retention was minimal.

However, several limitations persist. Complex syntactic variations in morphologically rich Indic languages remain difficult to capture reliably. The selection and optimization of FAISS-based vector databases specific to Indic languages continues to pose a challenge, influencing both retrieval efficiency and quality. Additionally, the BLOOMZ model currently supports a limited subset of Indic languages, narrowing the system's broader applicability. Issues

**Table. 6.** Domain-wise Comparison of Ground truth, Zero-Shot Hallucinations, and proposed RAG based translations

| Domain | Original Sentence | Reference Sentence | Zero Shot Translation | RAGMT Translations |
|---|---|---|---|---|
| **Healthcare** | The patient showed mild symptoms. | रोगी में केवल हल्के लक्षण पाए गए। | पाटिएंट था zk!# हलक ंहैं संज ।। | रोगी में केवल हल्के लक्षण पाए गए। |
| **Legal** | The hearing is postponed. | मामले की सुनवाई टाल दी गई है। | सुनवाई #@*% कल 9 बजे ₹₹ दोष तय करना जरूरी है। | मामले की सुनवाई टाल दी गई है। |
| **Sports** | He scored a winning goal. | उसने निर्णायक गोल किया। | स्कोर किया@@ गोल बॉल में रनिंग रनिंग फिर झटका झटका। | उसने निर्णायक गोल किया। |
| **Education** | Exams start next week. | अगले सप्ताह से परीक्षाएं शुरू होंगी। | एक्ज़ाम फ़रवरी ##$$ 2024 में स्टार्टअप होगा 11ल45प। | अगले सप्ताह से परीक्षाएं शुरू होंगी। |
| **News** | Streets were flooded. | सड़कों पर जलभराव हो गया। | फ्लडिंग हुई... सड़क पर **सड़क पर** सससस... *** | सड़कों पर जलभराव हो गया। |

such as out-of-vocabulary (OOV) terms and inconsistent handling of domain-specific terminology affect translation robustness, highlighting the need for enhanced tokenization strategies. Future work may explore expanding the framework to more languages, adopting larger multilingual models, and integrating hallucination detection or low-resource adaptation techniques to boost reliability and real-time performance.

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